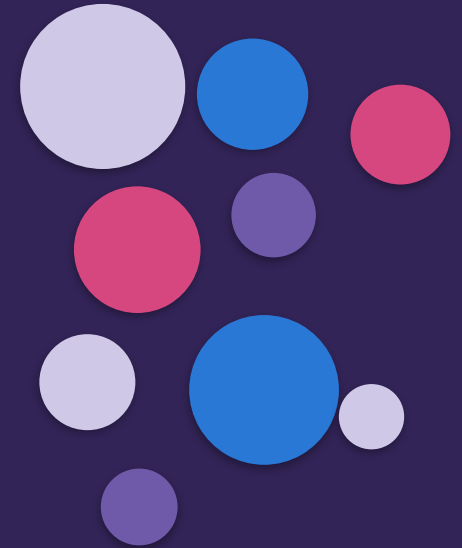


Nuclei Instance Segmentation on PanNuke

Progress report — Course Project 5
October 7, 2026 · bdd26 / jxh



- **The task**

- Find every nucleus in H&E images
- Separate the ones that touch
- Classify each into 1 of 5 types

~7,900 image patches

19 tissue types

5 nucleus classes

Dead (dying) nuclei: the hardest class

- **How we score: PQ**

higher = better · the official PanNuke metric

A prediction only counts when all three are right:

1

Found?

it is detected

2

Boundary right?

shapes overlap

3

Type right?

the class matches

Official 3-fold protocol

Last checkpoint

No leaked weights

Reproductions on PanNuke land at mPQ 0.46 – 0.50; every point is hard-won.

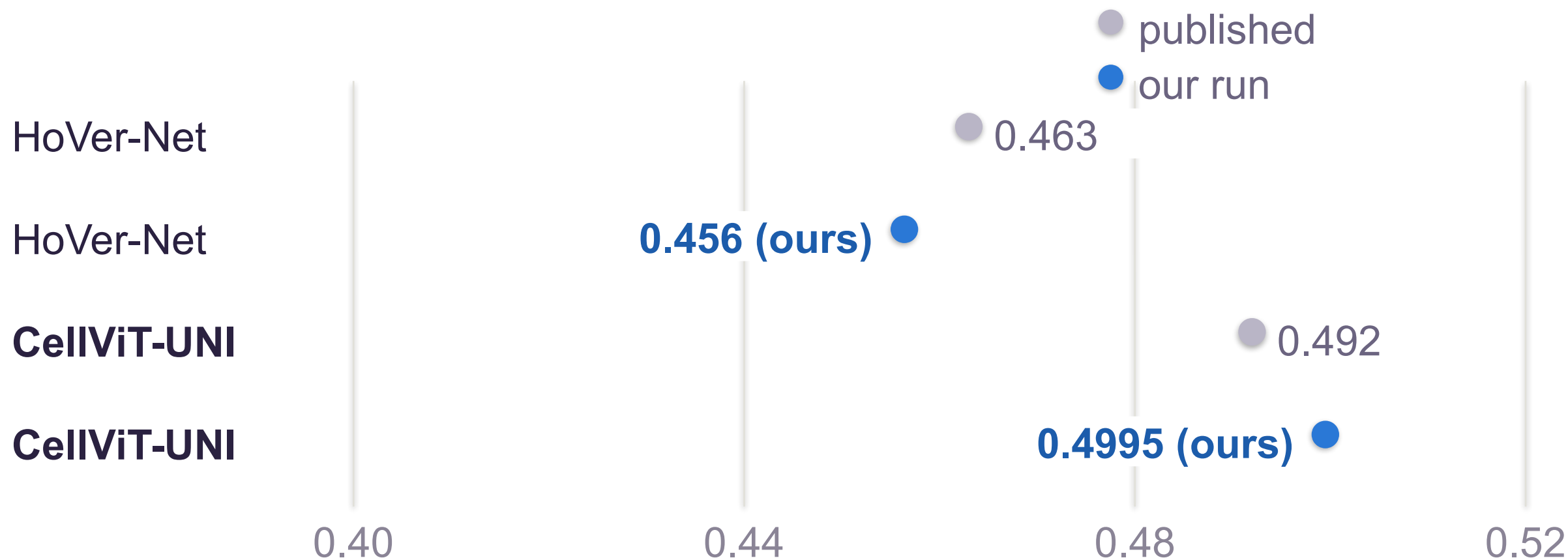
- **First: a pipeline you can trust**

- 1 **Compact data pipeline** — streamed from zip: 37 GB → 3 GB
- 2 **Exact evaluator** — bit-identical to the official metric code
- 3 **One strict stack** — 5 architectures, identical rules
- 4 **213 automated tests** — metrics, decoding, targets, plumbing

Every number in this deck comes from this pipeline.

- **Baselines reproduce the published numbers**

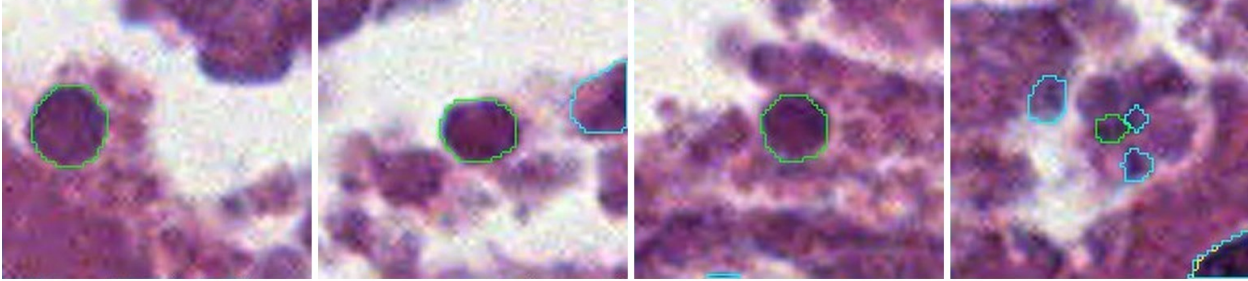
mPQ, 3-fold mean, no TTA



Both baselines land on the published numbers → the pipeline is sound.

- **The bottleneck: Dead nuclei are missed, not mistyped**

Dead missed (n=407)



Green: ground-truth Dead nuclei the model missed entirely

38.5%

of Dead nuclei missed entirely
(IoU < 0.1)

8 – 15%

missed rate, other classes

69% of matched Dead
are typed correctly

→ **a detection problem**

- **Stage 1 — three new-paradigm attacks**

Sep 25 – 28 · all judged against pre-registered rules

A

**Language-
anchored typing**

CONCH VLM re-types
nuclei from text

Null → diagnosis

B

**Failure-driven
synthesis**

diffusion makes hard
examples

Closed — no gain

C

**Counterfactual
stress test**

re-render images,
keep labels

Typing-only proxy

All three closed with informative negatives — and one key diagnosis.

- **Stage 1A — language anchoring (CONCH)**

the model can name Dead nuclei — it just doesn't find them

0.695

zero-shot typing accuracy (CONCH pooling)

0.88

zero-shot Dead recall

- **Fused into the model:** test mPQ unchanged (0.4951 → 0.4953)
- **Also null:** loss re-weighting · balanced sampling · recovery

Typing was never the bottleneck — detection is.

- **Stage 1B — generative synthesis (PixCell)**

make more hard examples with a tissue generator

- **Mined the failures:** missed Dead are small & isolated; merges sit in same-class clusters
- **Copy-paste real Dead nuclei:** hurts — Dead PQ -0.031
- **In-context diffusion (1,146 images):** no gain — mPQ 0.4900, inside seed noise
- **Structural blocker:** mask-only conditioning cannot steer appearance by class

Verdict: closed

Honest scope: these recipes failed — other data interventions stay open.

- **Stage 1C — counterfactual stress testing**

keep the labels, re-render the image → cheap robustness probe

- **Idea:** a generator as an evaluation simulator for stress testing
- **Result:** CF ranking vs real ranking is inverted ($\rho = -0.89$)
- **Mechanism:** label-fixed renders stress typing ($\rho +1.00$), never detection

Real architecture gaps live on the detection axis — a renderer that keeps labels fixed cannot touch it.

Verdict: typing stress test, not a transfer proxy

• Stage 2 — two attacks on the Dead deficit

Sep 28 – Oct 1 · detection-first architecture vs 2× resolution

Detection-first architecture

KongNet (CenterNet-style)

0.346 mPQ (−11 pts)

- touching same-class nuclei fuse
- interior-Dead miss: 0.50 vs 0.35

Negative

2× working resolution

512 px input & decoding

Dead ↑ at a priced cost

- interior-Dead miss 0.35 → 0.25
- cost: mPQ −0.009 · bPQ −0.007

Priced trade

Resolution helps Dead — can we keep the gain and drop the cost?

- **Stage 3 — making the trade free**

Oct 2 – 5 · four audited steps

1 Scale-aware decoding

fix decoder constants →
recovers ~70% of the cost

2 M/S lever

val-selected merge + fragment
fix — 9/9 runs

3 P2: gated additions

borrow 2× detections — Dead
↑, guard fails

4 EF-P2: 30px filter

keep credible additions — bPQ
tax ≈ 0

EF-P2 passes the pre-registered rule on 9 matched pairs.

P2 alone failed its bPQ guard (-0.0009); the filter removed exactly that tax.

- **Current best — EF-P2 vs matched baseline**

3-split mean over 9 matched (split, seed) pairs

Dead PQ — the target metric

baseline

0.173

EF-P2

0.180

+0.007 Dead PQ

0.00

0.10

0.20

0.4976 → 0.4983

mPQ (+0.0007)

0.6653 → 0.6653

bPQ ($\Delta \approx 0$ — the guard holds)

8 ↑ · 1 = · 0 ↓

per-pair Dead PQ signs

Candidate, not proof: CIs span zero · test folds reused across rounds

- **Running now — HV target threshold controls**

launched Oct 6 · the mechanism behind the 2× Dead gain

Is the 2× Dead gain about resolution — or about which tiny nuclei receive HV training signal?

Official HV targets ignore nuclei < 30 working pixels. At 2× that reaches smaller nuclei than at 1× — scale and supervision were confounded.

x1 · hv30

x1 · hv8

x2 · hv30

x2 · hv120

8 fresh runs

4 arms × 2 seeds, 130 epochs

No results yet

validation-only · test folds untouched

- **Why trust these numbers**

- **Pre-registered rules** — decision criteria fixed before any test data is read
- **Validation-only tuning** — each test fold is read once per round, last checkpoint
- **Multi-seed everywhere** — seed noise measured: mPQ ± 0.002 , Dead PQ ± 0.007
- **Honest negatives** — failed routes are reported as carefully as wins

Published ‘all-data’ checkpoints are never scored as test results.

Where we stand

- A strict, leak-free pipeline and reproduced baselines.
- An honest map of what does not work (stages 1–2).
- One robust gain: EF-P2, +0.007 Dead PQ, no bPQ cost.
- **The mechanism experiment is running now.**

Next: finish the 8 HV-control runs → audited analysis → freeze the candidate and validate it independently.

Thank you

